GABRIEL DA COSTA MERLIN

São Carlos, São Paulo, Brazil

Email ◊ LinkedIn ◊ GitHub ◊ Google Scholar

EDUCATION

Universidade de São Paulo (USP), ICMC **Jan 2022 - Dec 2026**
B.Sc. in Computer Science - GPA: 9.4/10 | Rank: 2nd/92

SKILLS

Programming: C/C++, Python
DS/ML: NumPy (4 years), Pandas (4 years), MatPlotlib (4 years), Scikit-Learn (4 years), PyTorch (3 years)
Languages: Portuguese (Native), English (TOEFL iBT C1)

RESEARCH EXPERIENCE

Foundation Models & Time Series | *Universidade de São Paulo / FAPESP* **Sep 2025 – Nov 2026**

- I developed TS2TabPFN, a novel framework combining time series feature extraction with tabular foundation models, achieving new state-of-the-art performance in extrinsic regression and matching top ensembles in classification, while achieving up to a 700x speedup on inference in select cases.

Neural Architecture Search for TSER | *Universidad de Granada / FAPESP* **Apr 2026 – Jun 2026**

- I constructed a 1D convolutional search space using FCN and ResNet primitives to empirically evaluate NAS strategies across 15 benchmark datasets, assessing predictive gains, computational efficiency, and energy consumption for time series extrinsic regression.

Structural Analysis of Time Series Models | *Universidade de São Paulo / FAPESP* **Jun 2024 – May 2025**

- I built end-to-end PyTorch pipelines to systematically evaluate modified FCN architectures, isolating the impact of structural parameters such as convolutional depth, predictor heads, and filter counts on time-series performance across diverse benchmark data tasks.

mHealth & Mobile Vision Systems | *Universidade de São Paulo / FAPESP* **Apr 2023 – Mar 2024**

- I engineered an end-to-end system for estimating clinical lab results from fingertip videos using a Flutter mobile app, OpenCV data processing, and FastAI deep learning models, successfully validating the feasibility of PPG video signals for automated clinical parameter estimation.

PUBLICATIONS (more details on Google Scholar)

- TS2TabPFN: Time Series Classification and Extrinsic Regression Through Feature Extraction and a Tabular Foundation Model **(ECML PKDD 2026)**
- Kaleidoscope: In-language exams for massively multilingual vision evaluation **(ICLR 2026)**
- FCKAN: Evaluating KAN for Time Series Classification and Extrinsic Regression **(ENIAC 2025)**

EXTRACURRICULAR ACTIVITIES

Rede de Avanço em IA (RAIA) — *AI Project Coordinator & Researcher*

- **BoneCheck** (Jan–Dec 2025): Coordinated an initiative for early osteoporosis detection in dental X-rays, building ensemble of vision models (ConvNeXt, Swin, EfficientNet), XGBoost, and VLMs.
- **BEMU** (Sep–Dec 2024): Scraped 10 years of entrance exams to build a multimodal high school dataset in Portuguese, directly contributing to an ICLR 2026 paper.

PET Computação — *Teaching Assistant & Instructor*

- **Overclock** (Jul–Dec 2023): Mentored undergraduate CS students in C programming.
- **Codifique** (Sep 2022–Jun 2023): Taught Python and basic game development to high school students.

DATA ICMC — *Member & Study Group Coordinator*

- **Study Groups & Ops** (Mar 2022–Feb 2025): Coordinated student-led study groups on ML, Deep Learning, and Time Series, while assisting in managing internal group operations.